ALICE L WALTON SCHOOL OF MEDICINE

Abdominal Wall Defects

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Adapted from: Hernias: Clinical Evaluation and Management

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate congenital abdominal wall defects based on embryology, anatomy, and clinical presentation.
2. Define and classify hernias and explain their pathophysiology and risk factors.
3. Compare and contrast direct and indirect inguinal hernias and femoral hern
4. Differentiate key clinical features of reducible, incarcerated, and strangulated hernias.

CASE CONNECTION

MR, 45-year-old male presents with a right groin bulge for one month, worsened by lifting heavy boxes at work. He reports mild discomfort but no severe pain. JR, a medical student, is asked to evaluate MR, describe his findings, consider possible diagnosis, and next management steps.

What features in MR's history and exam help define his condition? How does JR determine what the bulge is and guide MR's care? We'll revisit at the brick's end.

GO TO CONCLUSION ↓

Abdominal Wall (LO # 2)

The abdominal wall is a layered structure (skin, fascia, muscle, parietal peritoneum) that encloses the abdominal cavity, provides flexible support, protects internal organs, and regulates intra-abdominal function. (Fig. 1).

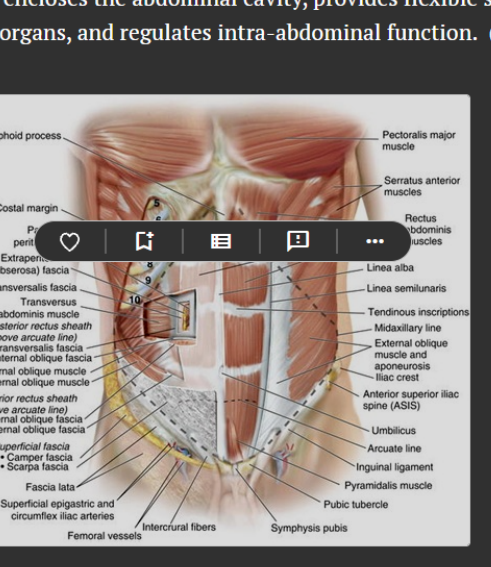


Figure 1: Anterior Abdominal Wall

Physiologically, the layers of muscle and fascia handle abdominal pressures ranging from resting, 0 to 5 mmHg (0.6 -0.7kPa) to peak pressures of 100 to 200 mmHg (13-26 kPa) during straining, such as defecation. Abdominal wall muscles, e.g. rectus abdominis, obliques, and transversus abdominis, contract to increase pressure. Pelvic floor muscles aid abdominal wall muscles in contraction during coughing; hence why patient's are asked to cough when palpating for inguinal hernias.

Loss of Abdominal Wall Integrity: Developmental Failure; Structural Weakness; Disruption

Loss of abdominal wall integrity means the muscles and connective tissue (fascia) of the abdomen are weakened, torn, or damaged where they no longer hold abdominal cavity contents inside the cavity. This may result in a visible protrusion, or bulge, and could involve massive amounts of abdominal contents. Can breakdown defects by 1) congenital abdominal wall defects 2) acquired abdominal d efects.

Congenital – Developmental Failure: Omphalocele; Gastroschisis; Umbilical Hernia (LO # 1)

Omphalocele (Fig. 2) is when an infant's intestines, liver, or other organs stick outside of the belly through the umbilical ring. The organs are covered in a thin, nearly transparent sac that usually is intact at birth. By the 10th or 11th week of pregnancy, the intestines normally go back into the abdomen; if not, an omphalocele occurs.

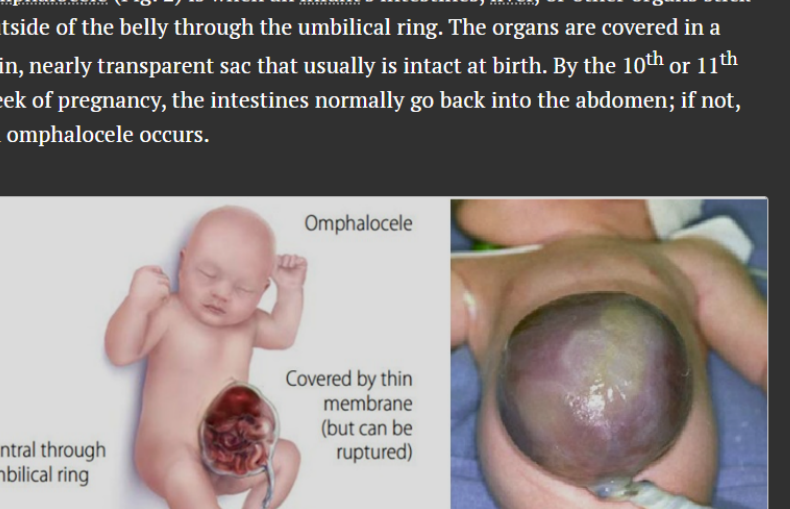

Omphalocele
Covered by thin membrane (but can be ruptured)
Central through umbilical ring

Figure 2: Omphalocele

The causes of omphalocele are mostly unknown. Reported associations include maternal tobacco/alc ancy, and pre-pregnancy obesity. Complications of an omphalocele may include: reduced/abnormal abdominal cavity size; infection especially if the sac is broken, or an organ becoming pinched or twisted. Babies born with an omphalocele often have other birth defects (e.g. heart defects, neural tube defects) and chromosomal abnormalities (Trisomy 18, 13, & 21 and chromosome 11 (11p15))

Gastroschisis

Gastroschisis (Fig 3) is a disorder where abdominal organs protrude (herniate) through a defect in the anterior abdominal wall adjacent to the umbilicus. The organs are not covered by a sac or skin. Gastroschisis arises in week 4 due to failure of fusion of the right and left anterior abdominal wall components in the midline during lateral folding. It is not associated with other congenital anomalies. Treatment is surgical replacement of the abdominal organs back into the abdominal cavity.

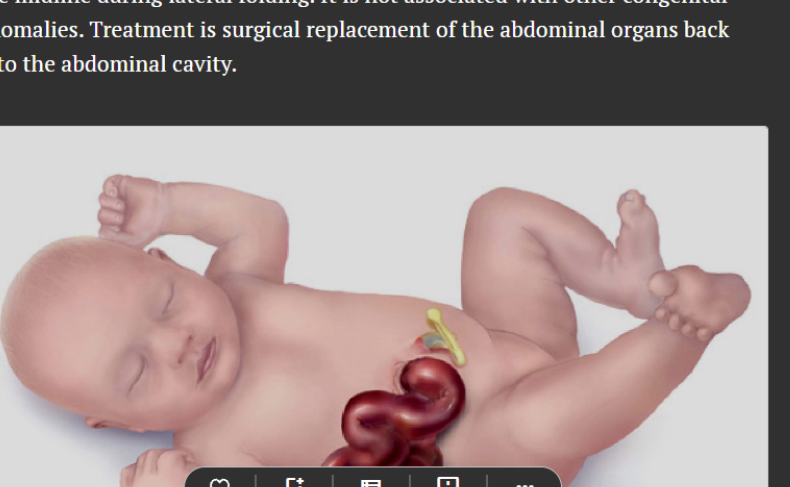
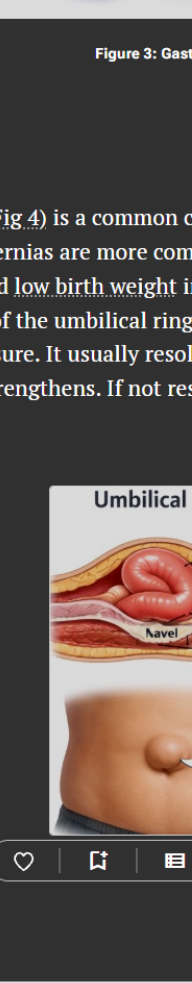


Figure 3: Gastroschisis

Umbilical Hernia

An umbilical hernia (Fig 4) is a common condition of infancy but may be seen in adults. Umbilical hernias are more common in people of African descent, premature infants, and low birth weight infants. These usually occur with failure or weakening of the umbilical ring closure, combined with increased intra-abdominal pressure. It usually resolves spontaneously by age 3 years as the abdominal wall strengthens. If not resolved by this time, surgical correction may be needed.



ACTIVE LEARNING QUESTION

A full-term male vaginal delivery neonate is born after an uncomplicated pregnancy with no prenatal care. Physical exam reveals a 3-cm midline abdominal wall defect at the base of the umbilical cord. The herniated abdominal contents are covered by a translucent membranous sac. The infant also has macroglossia and hypotonia. What is the most likely diagnosis?

- A. ○ Gastroschisis
- B. ● Omphalocele
- C. ○ Patent urachus
- D. ○ Umbilical hernia

Explanation

Correct Answer: (B) Answers not correct: (A) Gastroschisis: Free floating bowel loops without a covering sac; typically right sided.(C) Patent urachus: Presents with urine leakage from umbilicus, not herniated viscera. (D) Umbilical hernia: Covered by normal skin, not a translucent sac; usually appears after birth and

Hernia Definition and Classification (LO # 2)

A hernia is the protrusion of an organ or tissue through an abnormal opening in the structure that normally contains it, typically due to a weakness or defect in the muscular wall or fascia. Hernias most commonly occur in the abdomen, where increased intra-abdominal pressure can force contents through these weakened areas.

Clinically, hernias range from asymptomatic incidental findings to painful, potentially life-threatening conditions. Beyond physical effects, they can significantly impact quality of life, work capacity, and psychological well-being.

Key Structural Components

Every hernia has three essential components: the hernia sac, the hernia contents, and the defect or opening through which the herniation occurs. The hernia sac is a pouch-like structure formed by the peritoneum (the membrane lining the abdominal cavity) that protrudes through the defect. This sac creates a continuous lining between the abdominal cavity and the herniated contents, maintaining the integrity of the peritoneal space even as tissues displace abnormally. The hernia contents refer to the tissues or organs that protrude through the defect and into the hernia sac. These contents commonly include:

- Omentum: Mobile fold of visceral peritoneum that covers abdominal organs
- Small intestine: Most frequently herniated bowel segment
- Large intestine: Less commonly involved than small bowel
- Extraperitoneal fat: Particularly in certain hernia types (e.g. inguinal)
- Bladder: Occasionally seen in large inguinal or femoral hernias

The defect or opening represents the actual anatomical weakness through which tissues herniate. The size, location, and characteristics of this defect determine the hernia type. They can result from inherent tissue weakness, trauma, surgery, or increased pressure that gradually stretches and weakens the supporting tissues. The borders of the defect, known as the hernia margins, are critical landmarks during surgical repair.

Functional Impact

Hernias affect surrounding tissues through several mechanisms. Protrusion of tissues through a defect creates tension, causing discomfort or pain, which typically worsens with activities that increase intra-abdominal pressure (e.g., lifting, coughing, straining). Symptom severity depends on the hernia's size, location, and contents.

Complications is a key functional concern:

- Reducible: contents can be returned to the abdominal cavity
- Incarcerated (irreducible): contents are trapped, risking bowel obstruction
- Strangulated: compromised blood supply → ischemia and necrosis, requiring emergent surgery

Hernias may also cause chronic functional impairment. Patients often modify activities to avoid pain, leading to reduced mobility, work capacity, and quality of life. Large ventral h echanics, while inguinal hernias may formance. These impacts inform decisions about management and timing of intervention.

What are the three key structural components of a hernia?

What Are the Common Types and Anatomical Locations of Hernias? (LO # 2 & # 3)

Hernias are classified primarily by their anatomical location, with each type having distinct features that influence presentation, diagnosis, and management. While hernias can theoretically develop in many locations, certain sites are particularly susceptible due to inherent structural vulnerabilities or increased mechanical stress.

The most common wall hernia locations include the groin (inguinal and femoral) and abdominal wall (ventral: umbilical and incisional).

Inguinal and Femoral Hernias (L.O. # 3 / Fig 5)

Inguinal hernias are the most common abdominal wall hernias (~75%), occurring in the groin and more frequently in males due to anatomy related to testicular descent. They are classified as indirect or direct based on their relationship to the inguinal canal and inferior epigastric vessels:

- Indirect inguinal hernias pass through the internal inguinal ring, lateral to the inferior epigastric vessels, following the path of the spermatic cord (males) or round ligament (females). They result from a persistent processus vaginalis and are the most common type. The sac travels within the canal and may extend into the scrotum or labia majora. They occur at any age but are more common in children and young adults.
- Direct inguinal hernias protrude through a weakness in the posterior wall of the inguinal canal (Hesselbach's triangle), medial to the inferior epigastric vessels. They do not follow the canal, instead pushing directly through the abdominal wall. These are acquired, mostly seen in older adults due to muscle weakness, rarely extend into the scrotum, and have lower risk of incarceration.

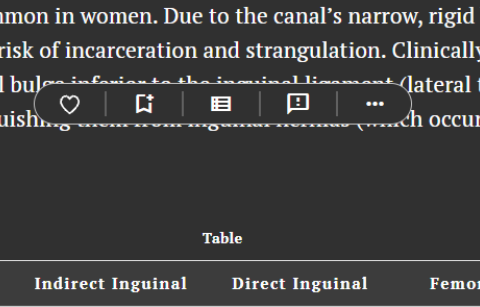


Figure 5: Hernias: Inguinal / Femoral

Key anatomical landmarks to distinguish between inguinal hernia types are:

- Inferior epigastric vessels: Define the lateral border of Hesselbach's triangle (Fig 6)
- Inguinal ligament: Forms the inferior border of the inguinal canal
- Pubic tubercle: Serves as a reference point for examination
- Internal inguinal ring: Entry point for indirect hernias

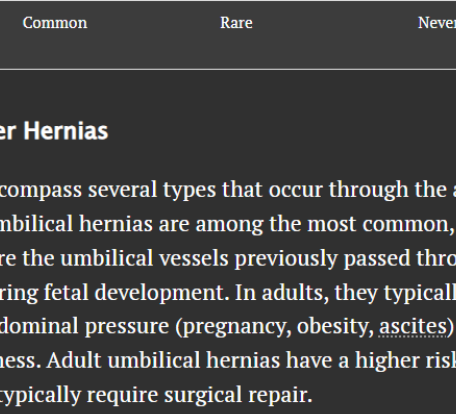


Figure 6: Hesselbach's Triangle

Femoral hernias protrude through the femoral canal (below the inguinal ligament, medial to the femoral vein). They account for ~5% of groin hernias and are more common in women. Due to the canal's narrow, rigid borders, they have the highest risk of incarceration and strangulation. Clinically, they present as a small bul (lateral to the pubic tubercle), distinguishi ch occur above the ligament).

Table

Feature	Indirect Inguinal	Direct Inguinal	Femoral
Anatomical Relation	Lateral to epigastric vessels	Medial to epigastric vessels	Below inguinal ligament
Etiology	Congenital or acquired	Acquired (tissue weakness)	Acquired
Gender Predominance	Males > Females	Males > Females	Females > Males
Risk of Strangulation	Moderate	Low	High
Extension to Scrotum/Labia	Common	Rare	Never

Ventral and Other Hernias

Ventral hernias encompass several types that occur through the anterior abdominal wall. Umbilical hernias are among the most common, occurring at the umbilicus where the umbilical vessels previously passed through the abdominal wall during fetal development. In adults, they typically result from increased intra-abdominal pressure (pregnancy, obesity, ascites) stretching a pre-existing weakness. Adult umbilical hernias have a higher risk of incarceration and typically require surgical repair.

Incisional hernias occur at prior surgical sites due to failure of proper abdominal wall healing, complicating ~10–20% of laparotomy incisions. Key risk factors include w , malnutrition, and technical issues with n size; large hernias may contain substantial intra-abdominal contents

Less common but clinically significant abdominal wall hernias include:

- Epigastric hernias: Occur midline between the xiphoid process and umbilicus
- Spigelian hernias: Protrude through the Spigelian fascia at the lateral edge of the rectus muscle
- Obturator hernias: Protrude through the obturator canal in the pelvis, most common in elderly, thin women

ACTIVE LEARNING QUESTION

A 67-year-old woman presents with a small, tender bulge below her right inguinal ligament that appeared after heavy lifting. Which type of hernia is most likely?

- A. ● Femoral hernia
- B. ○ Indirect inguinal hernia
- C. ○ Direct inguinal hernia
- D. ○ Umbilical hernia
- E. ○ Spigelian hernia

Explanation

Correct answer is Femoral hernia (A). Key finding: location is below the inguinal ligament, which is characteristic of femoral hernias. These hernias are more common in women and have a high risk of incarceration. Indirect inguinal hernias (B) and direct inguinal hernias (C) appear above the inguinal ligament. Umbilical hernias (D) occur at the umbilicus. Spigelian hernias (E) occur at the lateral edge of the rectus muscle.

What Is the Pathophysiology and What Are the Risk Factors for Hernia Development? (LO # 2)

Hernias result from an imbalance between intra-abdominal pressure and abdominal wall strength, allowing contents to protrude through a point of weakness. This process is typically gradual, as tissues progressively stretch and weaken over time, rather than occurring as an acute event (except with trauma).

Mechanisms of Herniation

The primary mechanism of hernia formation involves the interaction between tissue defects and pressure gradients. Tissue defects can be congenital or acquired and represent the anatomical point of vulnerability through which herniation occurs. Congenital defects include persistent processus vaginalis (predisposing to indirect inguinal hernias), umbilical defects, or inherent collagen disorders af ts commonly result from surgical incisions, trauma, or progressive weakening of supportive tissues due to aging, obesity, or repetitive strain.

Pressure dynamics play a crucial role in hernia development and progression. Normal activities such as coughing, lifting, and straining transiently increase intraabdominal pressure, and when pressure is repeatedly directed toward a weakened area, tissues stretch and the defect enlarges. Hernias often begin as reducible bulges that appear with straining and resolve at rest but may progress to persistent protrusions as tissue elasticity is lost.

At a structural level, impaired connective tissue integrity—often related to abnormal collagen composition (particularly decreased type I to type III collagen ratios) - reduces tissue strength and impairs wound healing. Inflammatory processes and matrix metalloproteinases also contribute to tissue degradation and remodeling, further compromising structural integrity.

The progression of hernia development typically follows these stages:

1. Initial tissue weakness or defect formation
2. Periodic bulging with increased intra-abdominal pressure
3. Stretching and enlargement of the fascial defect
4. Formation of a peritoneal sac protruding through the defect
5. Gradual herniation of abdominal contents into the sac

Contributing Risk Factors

Hernia development is multifactorial, typically resulting from a combination of structural weakness and increased mechanical stress on the abdominal wall. Risk factors can be broadly grouped into:

- Patient-specific (ge)
  - Advancing age reduces connective tissue strength and elasticity, increasing susceptibility to hernias—especially direct inguinal and incisional types
- Physiologic stressors (↑ intra-abdominal pressure from body habitus)
  - Obesity: ↑ intra-abdominal pressure → increased risk of ventral and incisional hernias
  - Significant weight loss: loss of supportive adipose tissue may unmask or predispose to herniation
- Pathologic conditions (impaired healing or tissue strength)
  - Malnutrition/cachexia: impaired collagen synthesis and wound healing → weakened abdominal wall and ↑ risk across all hernia types

Genetic predisposition also contributes, with family history and connective tissue disorders such as Ehlers-Danlos syndrome, Marfan syndrome, and osteogenesis imperfecta increasing susceptibility due to abnormal collagen integrity.

Common risk factors for hernia development include:

- Chronic cough (COPD, asthma, smoking)
- Straining: constipation or prostatism (voiding)
- Heavy lifting and physical labor
- Pregnancy and multiple births
- Previous abdominal surgery
- Ascites and peritoneal dialysis
- Diabetes mellitus impairs wound healing and tissue integrity

Surgical factors are relevant for incisional hernias, as wound closure technique and postoperative complications—wound infection, seroma formation, or wound dehiscence—s se hernia risk.

ACTIVE LEARNING QUESTION

A 62-year-old man with COPD develops a direct inguinal hernia. Which factor most likely contributed to his hernia development?

- A. ○ Advanced age
- B. ○ Family history
- C. ● Chronic cough
- D. ○ Diabetes mellitus

Explanation

The correct answer is Chronic cough (C). Chronic cough, a common symptom of COPD, causes repeated increases in intra-abdominal pressure that stress the abdominal wall. Advanced age (A) is a risk factor but less specific to this case. Family history (B) primarily influences indirect inguinal hernias. Diabetes (D) primarily affects wound healing and incisional hernias.

How Do We Perform Clinical Examination to Identify and Differentia

Clinical examination remains the cornerstone of hernia diagnosis, with imaging serving as a complementary tool. The examination combines visual inspection, palpation, and specific maneuvers designed to elicit characteristic findings for different hernia types. The goals of hernia examination include confirming the presence of a hernia, determining its anatomical type and location, assessing for complications (incarceration or strangulation), and identifying relevant factors that will impact management decisions.

Inspection and Palpation

Inspect the affected area at rest and with increased intra-abdominal pressure, such as coughing or straining. Examine the patient in both supine and standing positions, since some hernias are only apparent when upright or during Valsalva. For groin hernias, inspect the inguinal and femoral regions, comparing both sides for asymmetry or bulging. For ventral hernias, inspect the entire abdominal wall, with particular attention to the umbilicus and any prior surgical scars.

Palpation should be performed with warm hands and a gentle technique, especially if incarceration or strangulation is possible. Identify the location, size, and borders of any defect, determine whether a hernia sac is present, and assess its contents if possible. For groin hernias, palpate along the course of the inguinal canal, from the internal toward the external ring.

Key elements to assess during inspection and palpation include:

- Size and extent of the hernia bulge
- Exact anatomical location relative to landmarks
- Tenderness or pa
- Skin changes suggesting complications (erythema, edema)
- Consistency of the hernia contents (soft, firm, tense)

Inguinal hernias - the relationship of the bulge to the pubic tubercle and inguinal ligament helps with classification. Place a finger over the pubic tubercle, trace laterally along the inguinal ligament, and ask the patient to cough or strain.

- Indirect inguinal hernia: bulge follows the inguinal canal and may extend toward the scrotum
- Direct inguinal hernia: broader bulge that pushes directly anteriorly
- Femoral hernia: bulge located below the inguinal ligament and lateral to the pubic tubercle

You will need to assess the umbilical ring for an umbilical hernia defect while the patient is relaxed and while straining. For an incisional hernia, examine the full length of the surgical scar, as there may be multiple defects. Document the size, location, and relationship of each defect to the original incision.

Special Maneuvers

Several specific examination techniques help differentiate hernia types and assess their characteristics. The cough impulse test is fundamental to hernia examination. With the examiner's fingers placed over the suspected hernia site, the patient is asked to cough. A palpable impulse against the examiner's fingers suggests a hernia at that location. This test is particularly useful for small or subtle hernias that might not be immediately visible.

Reducibility assessment is crucial for determining hernia status and urgency. With the patient relaxed (usually supine), gentle pressure is applied to attempt to return the herniat A hernia may be completely reducible (contents return easily), partially reducible (some contents return), or irreducible (contents cannot be returned). Irreducibility raises concern for incarceration or strangulation, particularly if accompanied by pain or tenderness.

The silk glove sign helps differentiate direct from indirect inguinal hernias. After reducing the hernia, the examiner places a finger over the internal inguinal ring (approxim. 2 cm above the inguinal ligament midpoint) and asks the patient to strain. If the hernia reappears despite internal ring occlusion, it suggests a direct hernia. If the hernia is prevented by this maneuver, an indirect hernia is more likely.

Special examination maneuvers for specific hernia types include:

- Valsalva / Coughing: Patient to stand and bear down (or cough), which increases abdominal pressure
- Finger Invagination (Males): uses the index finger to invaginate the skin of the scrotum up to the superficial inguinal ring
- Deep Ring Occlusion: patient is supine and the hernia is reduced, the examiner compresses the deep inguinal ring (mid-inguinal point).

For suspected femoral hernias, the examiner should palpate below the inguinal ligament while the patient is standing. Femoral hernias can be difficult to detect due to their often small size and deep location. The examiner should place one hand above the inguinal ligament and the other below, asking the patient to cough while feeling for an impulse passing beneath the ligament. Comparison with the contralateral side is particularly important for subtle findings.

Where physical examination findings are equivocal or when examining obese patients where landm maging studies are considered: Ultrasound offers a dynamic, radiation-free option; CT scanning provides assessment of complex or recurrent hernias; MRI is occasionally used for complex.

The silk glove sign (occluding the internal ring), assessment of the hernia's relationship to the inferior epigastric vessels, and determining if the hernia follows the inguinal canal or protrudes directly forward.

Complications (LO # 4)

Complication (Fig. 7) assessment is priority in hernia evaluation, as it directly influences the urgency of intervention. Incarceration occurs when herniated contents become trapped outside the defect and cannot be reduced. While not immediately life-threatening, incarceration significantly increases the risk of progression to strangulation. Clinical signs of incarceration include irreducibility, pain or tenderness at the hernia site, and potentially symptoms related to the trapped or obstipation if bowel is involved).

Strangulation represents a surgical emergency and occurs when the blood supply to the herniated tissues is compromised, leading to ischemia and potential necrosis. Without prompt intervention, strangulated contents (particularly intestine) can become gangrenous, leading to perforation, peritonitis, sepsis, and potentially death. Clinical indicators of strangulation include severe pain, erythema or discoloration over the hernia, systemic signs of inflammation (fever, tachycardia), leukocytosis, and signs of bowel obstruction if intestine is involved.

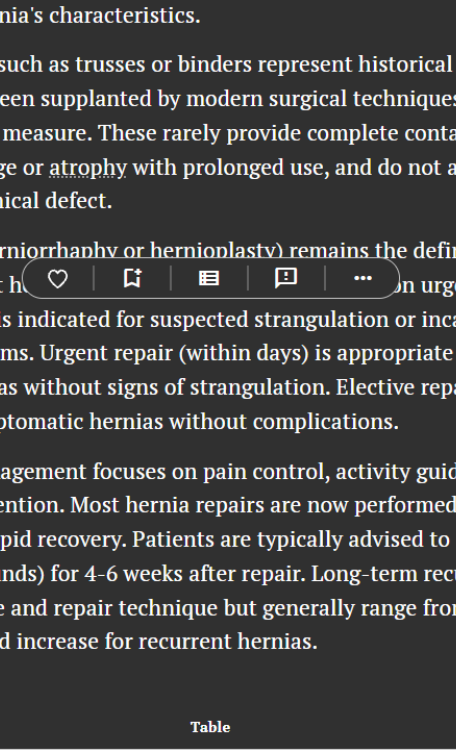

Skin
Abdominal wall weakens
Intestine
Peritoneum
Intestine pushes into sac
Intestine becomes trapped
Intestine becomes strangled

Figure 7: Progression of Hernia to Complication(s)

Femoral hernias have the highest risk of incarceration and strangulation (approximately 30-40% lifetime risk), followed by indirect inguinal hernias (approximately 10% risk), with direct inguinal hernias having the lowest risk among groin hernias. For ventral hernias, the risk correlates with the size of the neck of the herni necked hernias with large contents are at higher risk. Patient factors such as advanced age, comorbidities affecting tissue perfusion, and conditions causing increased intra-abdominal pressure also influence complication risk.

Signs and symptoms suggesting hernia complications include:

- Sudden onset or worsening of pain
- Irreducibility of previously reducible hernia
- Erythema, warmth, or discoloration over the hernia site
- Systemic signs: fever, tachycardia, hypotension
- Nausea, vomiting, abdominal distention (with bowel involvement)
- Absent bowel sounds or signs of peritonitis in advanced cases

Treatment Strategies

Hernia treatment approaches span from conservative management to emergency surgery. Non-surgical management may be appropriate for asymptomatic or minimally symptomatic hernias in patients with significant comorbidities or limited life expectancy. This "watchful waiting" approach involves patient education about warning signs of complications, activity modifications to reduce symptom exacerbation, and regular monitoring for changes in the hernia's characteristics.

External supports such as trusses or binders represent historical approaches that have largely been supplanted by modern surgical techniques and should be only temporary measure. These rarely provide complete containment, may cause tissue damage or atrophy with prolonged use, and do not address the underlying anatomical defect.

Surgical repair (herniorrhaphy or hernioplasty) remains the definitive treatment for most h n urgency. Emergency repair is indicated for suspected strangulation or incarceration with significant symptoms. Urgent repair (within days) is appropriate for incarcerated hernias without signs of strangulation. Elective repair can be scheduled for symptomatic hernias without complications.

Postoperative management focuses on pain control, activity guidance, and complication prevention. Most hernia repairs are now performed as outpatient procedures with rapid recovery. Patients are typically advised to avoid heavy lifting (>10-15 pounds) for 4-6 weeks after repair. Long-term recurrence rates vary by hernia type and repair technique but generally range from 1-10% for primary repairs and increase for recurrent hernias.

Table

Clinical Presentation	Management Approach	Timing
Asymptomatic, low-risk hernia	Watchful waiting or elective repair	Non-urgent
Symptomatic, reducible hernia	Elective surgical repair	Weeks to months
Incarcerated, painful hernia	Urgent surgical repair	Days
Signs of strangulation	Emergency surgical repair	Immediate
High surgical risk, minimal symptoms	Conservative management	Ongoing monitoring

ACTIVE LEARNING QUESTION

A 78-year-old m reducible, mildly symptomatic direct inguinal hernia. What is the most appropriate initial management?

- A. ○ Emergency surgical repair
- B. ○ Watchful waiting only
- C. ● Elective repair if medically optimized
- D. ○ Truss application
- E. ○ Urgent repair within 24 hours

Explanation

Correct answer is Elective repair if medically optimized (C). For this elderly patient with significant comorbidities but a symptomatic hernia, elective repair after medical optimization offers the best balance of risks and benefits. Emergency surgery (A) is not indicated without signs of complications. Watchful waiting (B) is reasonable but less optimal given the symptoms. Truss application (D) is generally ineffective and may cause complications. Urgent repair (E) is unnecessary without incarceration or strangulation.

CASE CONNECTION K TO INTRODUCTION ↑

MR's right groin bulge that worsens with exertion (e.g., heavy lifting) is classic for an inguinal hernia. His age (45) and male sex favor an inguinal over a femoral hernia. Mild discomfort without severe pain suggests a reducible, non-strangulated hernia. His history of heavy lifting is a key risk factor due to increased intra-abdominal pressure.

Summary

- A hernia is the protrusion of an organ or tissue through an abnormal opening and consists of three key components: the hernia sac (peritoneal pouch), hernia contents (protruding tissue or organs), and the defect (anatomical weakness).
- Hernia pathophysiology reflects an imbalance between mechanical forces and tissue integrity, with contributing risk factors including age, sex, obesity, genetic predisposition, chronic cough, constipation, and prior surgery.
- Inguinal hernias are classified as indirect (passing through the internal inguinal ring lateral to the inferior epigastric vessels) or direct (protruding through Hesselbach's triangle medial to the inferior epigastric vessels), whereas femoral hernias occur below the inguinal ligament and carry the highest risk of strangulation.
- Management decisions balance surgical and non-surgical approaches based on hernia characteristics, patient factors, and complication risk, with strangulated hernias requiring emergency intervention while asymptomatic hernias may be managed conservatively in select patients.

Review Questions

1. What are the three key structural components of a hernia?

- A. ○ Sac, contents, and defect
- B. ○ Muscle, skin, and fat
- C. ○ Ligament, sac, and ligament
- D. ○ Bone, peritoneum, and muscle

Explanation (requires correct answer)

2. Which hernia type is most common in males and occurs lateral to inferior epigastric vessels?

- A. ○ Direct inguinal hernia
- B. ○ Indirect inguinal hernia
- C. ○ Femoral hernia
- D. ○ Umbilical hernia
- E. ○ Spigelian hernia

Explanation (requires correct answer)

3. What is the primary pathophysiological mechanism in hernia formation?

- A. ○ Sudden trauma to peritoneum
- B. ○ Complete tissue regeneration
- C. ○ Pressure exceeds tissue strength
- D. ○ Infection causing wall rupture
- E. ○ Muscle hypertrophy over defect

Explanation (requires correct answer)

4. Which clinical maneuver helps distinguish direct from indirect inguinal hernias?

- A. ○ Silk glove sign
- B. ○ Transillumination test
- C. ○ Cough impulse test
- D. ○ Valsalva maneuver
- E. ○ Zieman's technique

Explanation (requires correct answer)

5. A 62-year-old man presents with a painful right groin bulge that is acutely tender and irreducible over the past 8 hours. He reports nausea but no vomiting. On exam, the mass is firm, non-reducible, and mildly tender and no skin changes. Bowel sounds are present. Which is most appropriate condition?

- A. ○ Femoral lymphadenopathy
- B. ○ Incarcerated hernia
- C. ○ Direct hernia
- D. ○ Strangulated hernia

Explanation (requires correct answer)

How would you rate this Brick?

References

- Sabiston Textbook of Surgery. The Biological Basis of Modern Surgical Practice. chapter 45

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☐ Differentiate congenital abdominal wall defects based on embryology, anatomy, and clinical presentation.
- 02 ☐ Define and classify hernias and explain their pathophysiology and risk factors.
- 03 ☐ Compare and contrast direct and indirect inguinal hernias and femoral hernias.
- 04 ☐ Differentiate key clinical features of reducible, incarcerated, and strangulated hernias.

Objective Confidence: 0/4